

in the same axis, like for a row. For example:

```
deltaA = A[1: , :] - A[:,1: ,:]
```

Indices are zero-relative, so the first term means a slice of the array starting with the 2nd row, and all the columns. The second term represents a slice of the array starting with the 1st row till the last-but-one row, and all the columns. Both slices are of the same size, and can thus be subtracted. You get speed as well, because the element-wise operations are done by the C library.

Now, let's assume that you wish to convolute an image with a 3x3 mask, M. This operation corresponds to element-by-element multiplication of the values in a 3x3 window around a pixel with corresponding mask elements, and then summing the result. For the sake of simplicity, you may ignore the border elements.

```
NR, NC = A.shape
Result = [ [ (M*A[i-1:i+2, j-1:j+2]).sum()
             for j in range(1, NC-1)]
```

```
for i in range(1, NR-1)]
```

You can use the Laplacian mask to get another image of the edges.

You will find a rich set of routines for filtering, morphology, segmentation and feature extraction in the *ndimage* module. To know more, just look at the section on image processing, analysis and face recognition (an example of machine learning) at <http://scipy-lectures.github.com/advanced/>.

New applications in image processing are waiting to be created. One wishes that Matlab would be replaced by SciPy in colleges—only then is it likely that a clever student will create a killer app. 

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